

Seth Ricks

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EDUCATION

Bachelor's in Electrical Engineering

Anticipated: December 2026

Brigham Young University - Idaho

- Overall GPA: 4.0
- Scholarships: Brigham Young University-Idaho Academic Grant

RELEVANT WORK EXPERIENCE

Control Systems Intern

May 2026 - July 2026

Beyond New Horizons

- Created complex LabVIEW auxiliary trigger system for wind tunnel imaging
- Programmed low-level cRIO/FPGA VI logic to generate 30 precise, configurable waveforms simultaneously
- Authored comprehensive 30-page technical document to guide system operation and user modifications

Research Assistant Intern

May 2025 - July 2025

FRoSt LAB

- Built a PyQt/ROS GUI to monitor autonomous underwater vehicles, supporting a 10-person team in field tests
- Presented GUI updates to lab, team, and campus groups; authored a 26-page GitBook documenting project
- Collaborated with team and reported to leaders weekly, ensuring task completion and smooth project workflow

Drafting Assistant Intern

January 2025 - April 2025

Engineered Systems Associates

- Designed 5 engineering documents on average per week using AutoCAD and BlueBeam
- Delivered quality work under tight weekly deadlines in a high-pressure environment
- Averaged 3 hours/day of AutoCAD training during slow periods, improving efficiency in peak workloads

PERSONAL PROJECTS

Interactive Dueling Wands

September 2025 - December 2025

- Wrote code in C++ for ESP-NOW communication, LCD displays, and game logic
- Modeled and 3D printed wand and base station parts with CAD software
- Created 2-layer PCB for easier circuit integration and reproduction of wands

Autonomous Chess Board

May 2025 - September 2025

- Implemented homing sequence and tracked motor positions using 2 quadrature rotary encoders
- Developed and 3D-printed a dual motor gear system with CAD software
- Calculated linear-to-polar coordinate conversion for robotic movement control with 1 degree precision

TECHNICAL SKILLS

Programming Languages: Python, C#, C, C++, LabVIEW, Verilog, ARM assembly, (in order of proficiency)

Software/IDEs: ROS, SolidWorks, AutoCAD, LTSpice, MATLAB, Logisim, KiCad, Raspberry Pi, STM32Cube, Arduino

Hardware Skills: PCB Design, Soldering, Debugging, Circuit Analysis

English (Native), Spanish (Intermediate)